

Maresh C. Gandikota

CONTACT INFORMATION

H108, Academic Block, ICTS
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EDUCATION

Ph.D., Physics, Syracuse University, Syracuse, NY, USA **2021**

Thesis Title: Interplay of Geometry and Mechanics:
Disordered Spring Networks and Shape-Changing Cerebella
Thesis Advisor: Prof. J. M. Schwarz

Integrated M.Sc., Physics, National Institute of Science Education
and Research, Odisha, India **2015**

Thesis Title: Martensitic phase transition of bacteriophage T4's tail sheath
Thesis Advisor: Prof. Somendra Bhattacharjee
G.P.A.: 8.2

Twelfth grade board exam, National College, Bangalore, India **2010**

Score obtained: 81%

Tenth grade board exam, HJKP, Bangalore, India **2008**

Score obtained: 97.4%

EMPLOYMENT

Postdoc, International Centre for Theoretical Sciences, Karnataka, India **2024–Present**

Postdoc, Department of Chemistry, Columbia University, NY, USA **2021–2024**

Research Assistant, Department of Physics, Syracuse University, NY, USA **2016–2021**

Teaching Assistant, Department of Physics, Syracuse University, NY, USA **2015–2020**

PUBLICATIONS

10. A.D. Chen, [M. C. Gandikota](#), and A. Cacciuto, “The shape of ideal and self-avoiding ribbons: From polymers to surfaces”, *Soft Matter* (*Accepted*).
9. A.D. Chen*, [M. C. Gandikota*](#), and A. Cacciuto, “[The shape of cleaved tethered membranes](#)”, *Soft Matter*, 21.6 (2025).
* Equal contribution
8. [M. C. Gandikota](#), Shibamanda Das and A. Cacciuto, “[Spontaneous crumpling of active spherical shells](#)”, *Soft Matter*, 20.17 (2024).
7. [M. C. Gandikota](#) and A. Cacciuto, “[The crumpling transition of active tethered membranes](#)”, *Soft Matter*, 19.28 (2023).
6. [M. C. Gandikota](#) and A. Cacciuto, “[Rectification of confined soft vesicles containing active particles](#)”, *Soft Matter*, 19.2 (2023).
5. [M. C. Gandikota](#) and A. Cacciuto, “[Effective forces between active polymers](#)”, *Physical Review E*, 105, 034503 (2022).

4. [M. C. Gandikota](#), Amanda Parker and J. M. Schwarz, “Rigidity transitions in zero-temperature polygons”, *Physical Review E*, 106, 055003 (2022).
3. [M. C. Gandikota](#) and J. M. Schwarz, “Buckling without bending morphogenesis: Nonlinearities, spatial confinement, and branching hierarchies”, *New Journal of Physics*, 23, 063060 (2021).
2. [M. C. Gandikota](#), Katarzyna Pogoda, Anne van Oosten, T. A. Engstrom, A. E. Patteson, P. A. Janmey, J. M. Schwarz, “Loops versus lines and the compression stiffening of cells”, *Soft Matter*, 16(18), 4389-4406 (2020).
1. S. Paul, [M. C. Gandikota](#), “Fourier Transform of Electric Signal using Kundt’s Tube”, *Student Journal of Physics*, 6(2), 95-100 (2016).

Manuscripts under review

1. *Self-avoiding tethered surfaces are always flat*
A. D. Chen, [M. C. Gandikota](#), M.J. Kim and A. Cacciuto
Under review in *Physical Review Letters* published by the American Physical Society

Manuscripts in preparation

1. *The jammed phase of infinitely persistent active matter*
[M. C. Gandikota](#), Rituparno Mandal, Pinaki Chaudhuri, Bulbul Chakraborty and Chandan Dasgupta
2. *Entropy production gradients in active matter systems*
M. Kim, [M. C. Gandikota](#), and A. Cacciuto

ONGOING PROJECTS

1. *Approach to equilibrium in the Riesz gas* with Anupam Kundu and Abhishek Dhar
2. *The jammed phase of systems with non-reciprocal interactions* with Shaon Chakraborty and Rituparno Mandal
3. *Stress correlations in active jammed systems* with Rituparno Mandal, Pinaki Chaudhuri, Bulbul Chakraborty and Chandan Dasgupta

TEACHING

Recitations/Substitute Lecturer

PHY 216, General Physics II for honors and majors	Spring & Fall 2017, Spring 2018
PHY 212, General Physics II	Spring & Fall 2016
PHY 211, General Physics I	Fall 2015

Labs

AST 101, Our Place in the Universe	Fall 2020
PHY 102, Major Concepts of Physics II	Spring 2019, 2020

Grading/Substitute Lecturer

PHY 635, Physical Cell Biology	Fall 2018
PHY 360, Vibrations, Waves and Optics	Fall 2018

Guest Lecturer

CHEM G4221, Statistical Thermodynamics	Fall 2022
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MENTORING

Alexandra Brown	REU program at Syracuse University
Min Jae Kim and Albert Chen	Columbia University

LANGUAGES	Human: English, Telugu (mother tongue), Kannada (native), Hindi.	
	Programming: Bash, C++, Python, MATHEMATICA.	
	Markup: L ^A T _E X, Markdown.	
	Operating System: Linux, Windows.	
AWARDS	Postdoctoral fellowship at ICTS-TIFR	2024
	Summer Graduate Fellowship, Soft Matter Program, Syracuse University	2017
	Henry Levinstein Fellowship, Physics Department, Syracuse University	2016
	All India Rank 24, Graduate Aptitude Test in Engineering (GATE)	2015
	Summer research fellow, Indian Academies of Sciences	2012
	INSPIRE fellowship, Department of Science and Technology, Government of India	2010-15
CONFERENCE/ WORKSHOP	Invited Talks	
	Discussion meeting on glassy, disordered and non-equilibrium soft matter IMSc, Chennai	2025
	Azim Premji University, Bengaluru, India	2025
	ICTS-TIFR, Bengaluru, India	2024
	Eindhoven University of Technology, Eindhoven, Netherlands.	2021
	IMSc, Chennai, India	2021
	Fibrous Networks in Biology, University of Pennsylvania, Pennsylvania, USA.	2018
	Contributed Talks and Posters	
	Unifying Concepts in Glassy Physics (UCGP), RRI, Bengaluru, India.	2025
	CompFlu, IISc, Bengaluru, India.	2025
	10th Indian Statistical Physics Community Meeting , ICTS-TIFR, Bengaluru, India.	2025
	Gordon Research Seminar & Conference, New Hampshire, USA.	2023
	Northeast Complex Fluids and Soft Matter workshop, New York, USA.	2023
	APS March Meetings.	2017, 2018, 2019, 2020
	Soft Matter School, University of Massachusetts, Massachusetts, USA	2017
SERVICE TO PROFESSION	Science outreach, eclipse event for residents of Shivakote, ICTS-TIFR.	2025
	Lecturer & Tutor, Summer School for Women in Physics, ICTS-TIFR.	2024
	Experiment Leader, Girls Science Day, Columbia University.	April, Nov 2023
	Manuscript reviewer for Physical Review E	
	Organizer, Soft Matter Journal Club, Syracuse University.	2019-2020
	Moderator, APS Conference for Undergraduate Women in Physics, Syracuse University.	2016
	Volunteer & Photographer, Active and Smart Matter conference, Syracuse University.	2016
	Volunteer, Science Day, annual joint outreach program to high school students, NISER, IOP, Bhubaneswar.	2011, 2012, 2014
NON-ACADEMIC	President, Argentine Tango Club, Syracuse University.	Fall 2017 - Fall 2020

Editorial Board, Jignasa, annual magazine of NISER, Bhubaneswar.

2014

REFERENCES

J. M. Schwarz

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Department of Physics
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Syracuse, NY 13244, USA
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A. Cacciuto

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3000 Broadway, New York, NY 10027, USA
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Karnataka, 560012, India
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Pinaki Chaudhuri

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Theoretical Physics
The Institute of Mathematical Sciences
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Somendra M. Bhattacharjee

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Department of Physics
Ashoka University
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